

Project Title

Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau

Team ID : LTVIP2026TMIDS74374

Team Size : 4

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Team member : Vemula Rohini

Objective

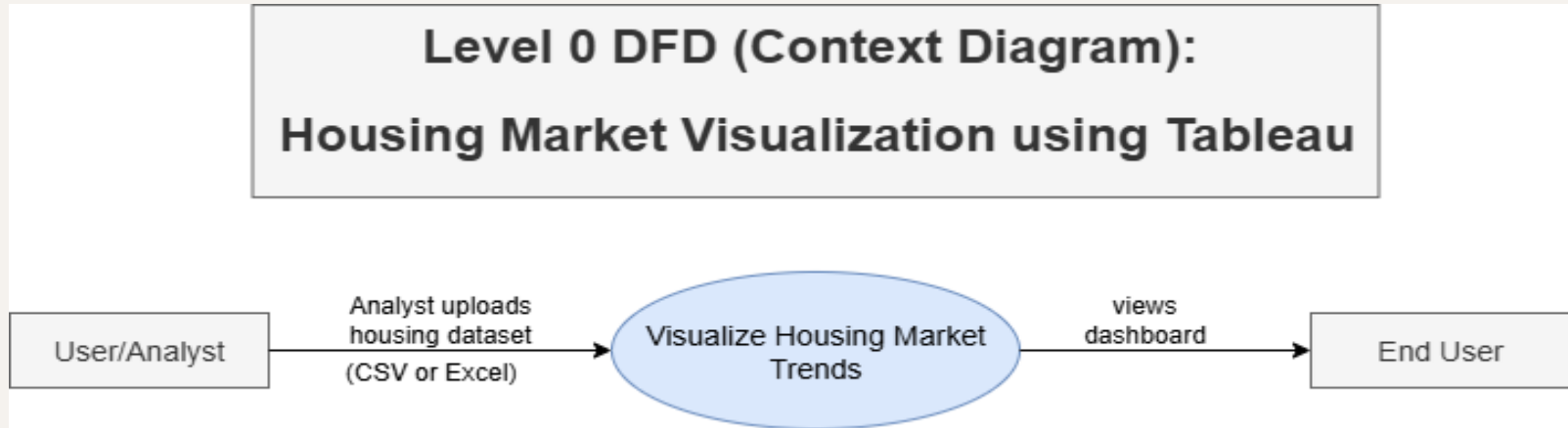
To analyze housing data and visualize trends such as house prices, age, and renovation impact using Tableau, enabling data-driven decision-making for stakeholders like real estate analysts and executives.

Dataset Used

- **Source:** kc_house_data.csv
- **Size:** ~21,000 records
- **Attributes:** Includes price, bedrooms, bathrooms, floors, condition, renovation year, square footage, and more.

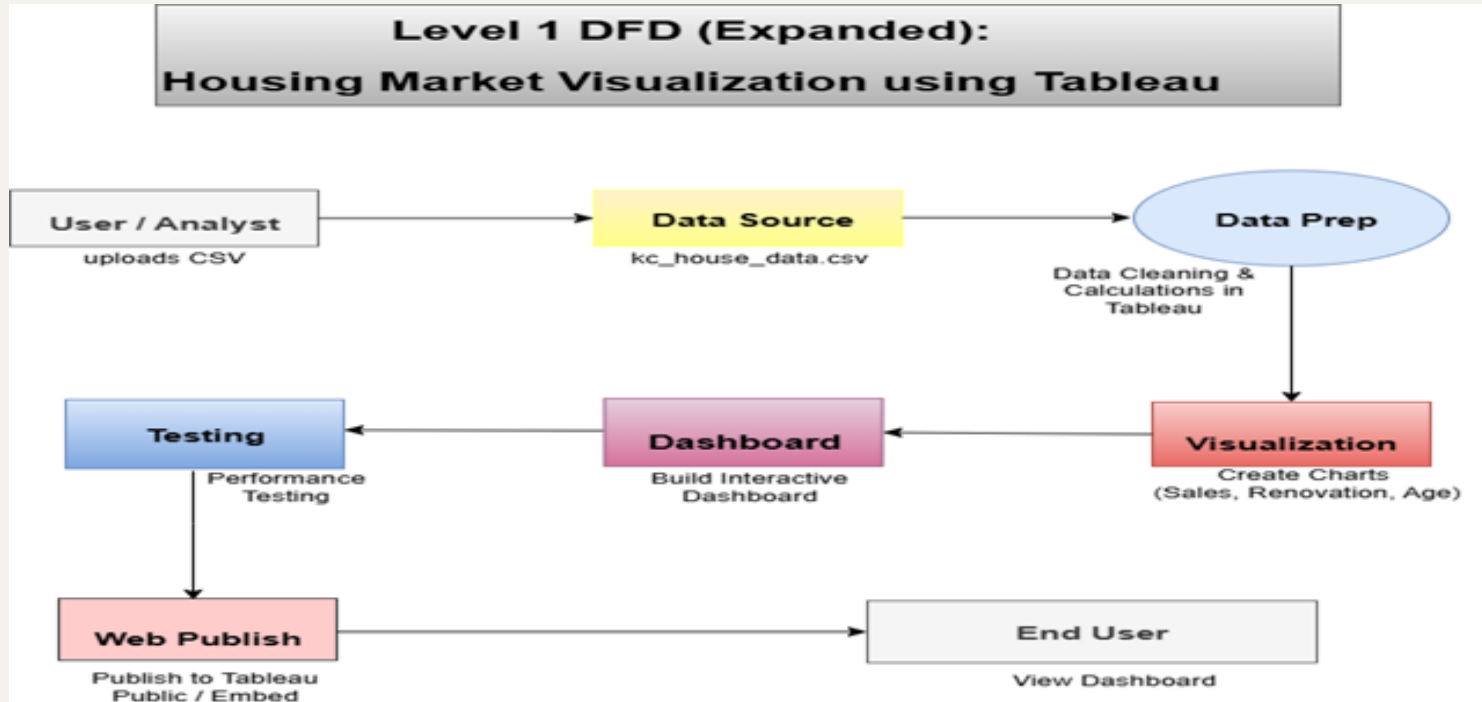
Data Flow Diagrams

DFD Level 0:



DFD Level 0 shows a high-level overview of the system where the User uploads housing data, the system processes it, and the End User views the visualized dashboard output.

DFD Level 1:



DFD Level 1 details the step-by-step flow from data upload to cleaning, visualization, dashboard creation, testing, and publishing for end-user access.

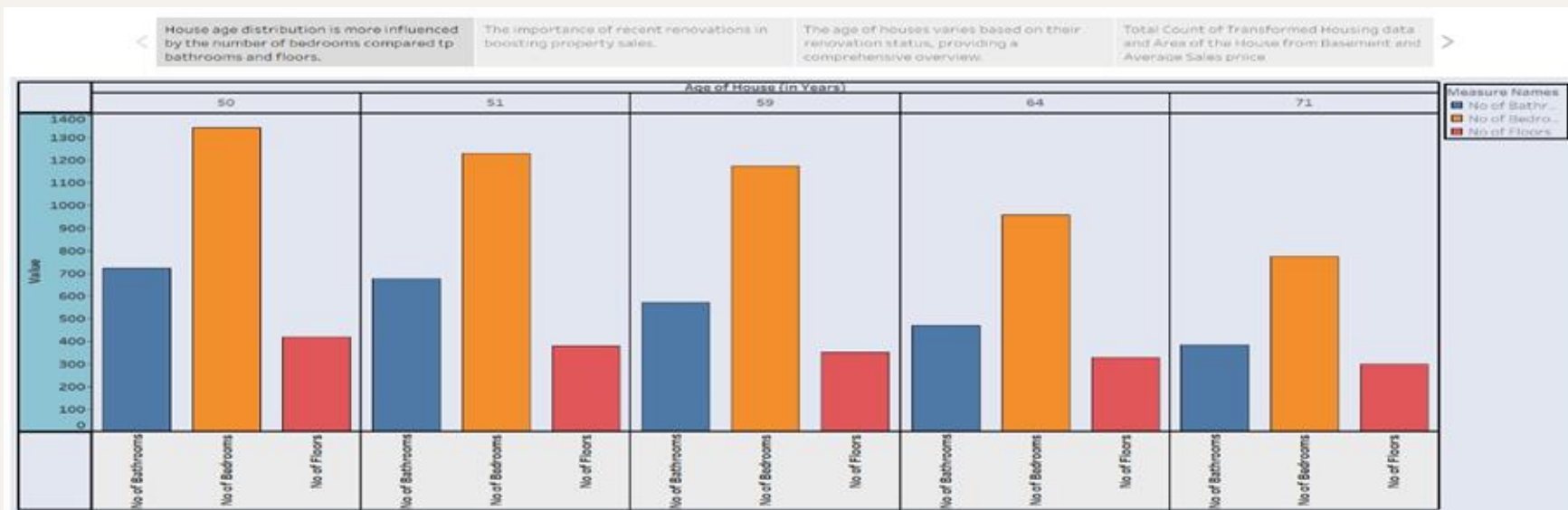
Data Preparation

- ❖ Cleaned the dataset by removing **null values** and **duplicates**
- ❖ Standardized column formats (e.g., **dates**, **numerical fields**)
- ❖ Created **calculated fields** like:
 - *House Age* = $2025 - yr_built$
 - *Years Since Renovation* = $2025 - yr_renovated$
- ❖ Renamed confusing or inconsistent column names for clarity
- ❖ Filtered irrelevant records (e.g., zero bedroom/floor houses)
- ❖ Ensured data is structured and ready for meaningful visualization in Tableau

Visualizations

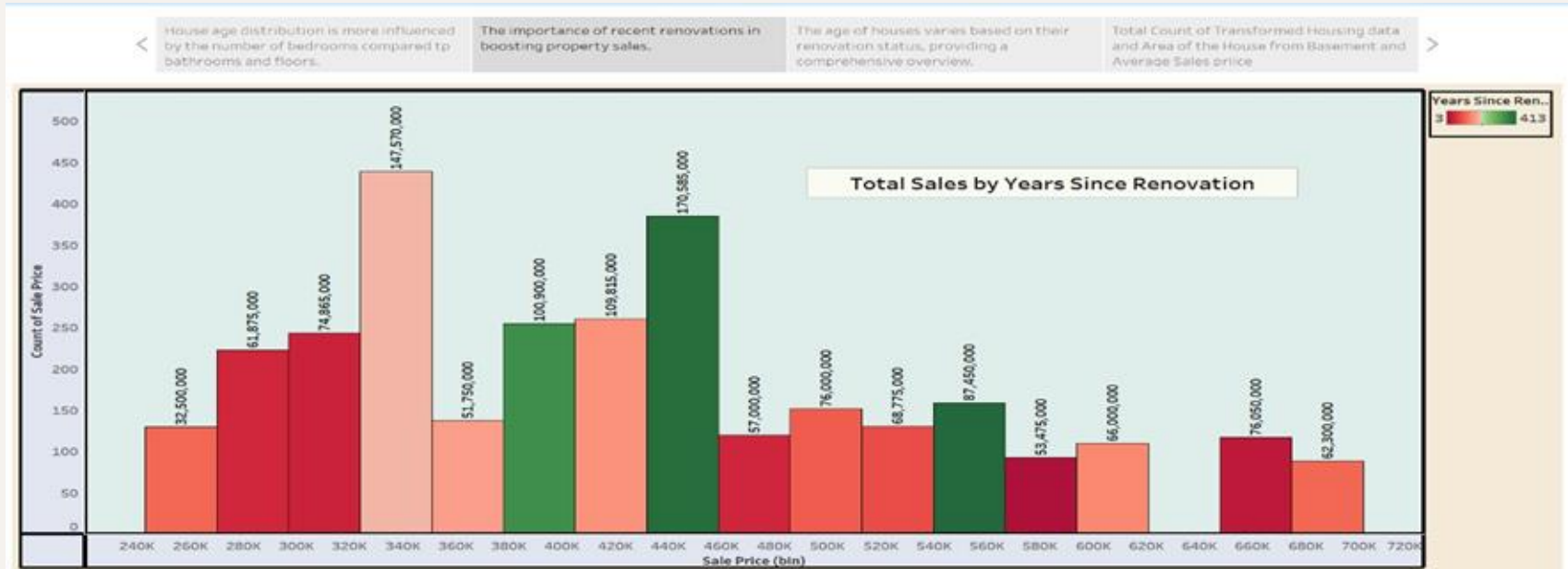
House Age vs Features

Displays how house age correlates with the number of bathrooms, bedrooms, and floors—highlighting which features influence housing age the most.



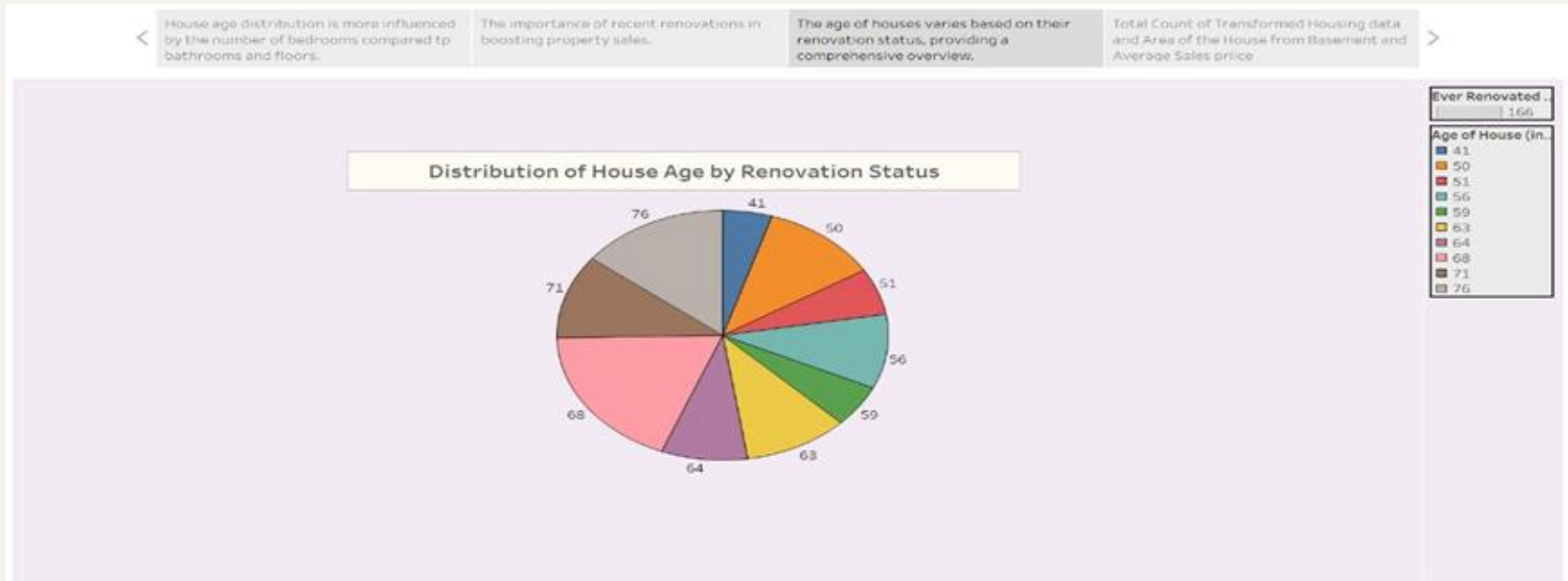
Total Sales by Years Since Renovation

Bar chart shows how recent renovations impact house sale prices—newly renovated homes have higher sales counts in specific price bins.



Distribution of House Age by Renovation Status

Pie chart illustrates the spread of house age grouped by whether they've been renovated, helping identify market trends across age categories.



Overall Dataset Summary

Summary metrics include:

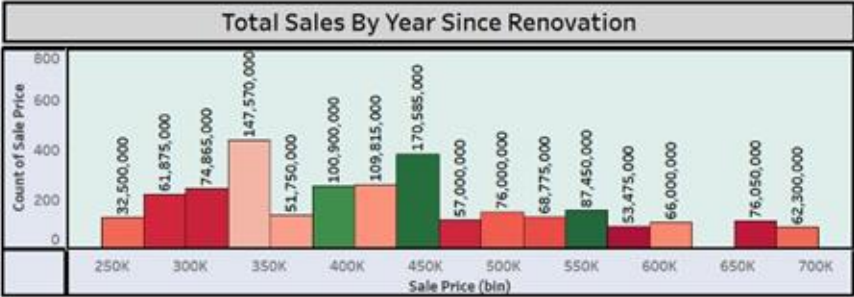
- Total data points analyzed: **21,609**
- Basement Area (sqft): **38,64,798**
- Average Sale Price: **₹511,619**

< House age distribution is more influenced by the number of bedrooms compared tp bathrooms and floors. The importance of recent renovations in boosting property sales. The age of houses varies based on their renovation status, providing a comprehensive overview. Total Count of Transformed Housing data and Area of the House from Basement and Average Sales price >	
Count of Transformed_Housing_Data2.csv	21,609
Area of the House from Basement (in Sqft)	38,643,798
Avg. Sale Price	511,619

Dashboard

Comprehensive Home Price Analysis

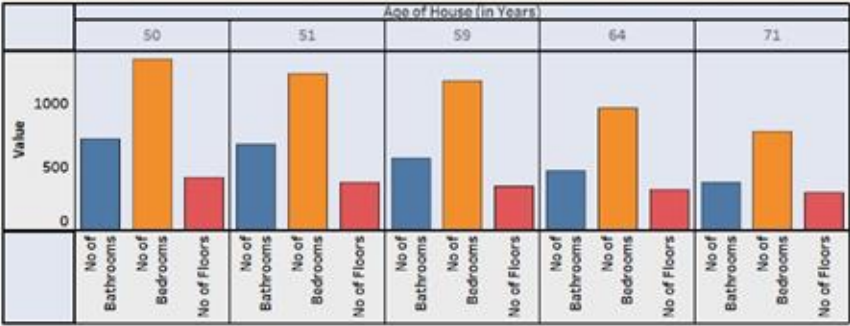
Count of Transformed_Housing_Data2.csv	21,609
Area of the House from Basement (in Sqft)	38,643,798
Avg. Sale Price	511,619



Distribution of House Age by Renovation Status



House Age Distribution by Number of Bathrooms, Bedrooms and Floors



Publishing

After creating the dashboard, it is published on **Tableau Public** for wider access.

This step explains how users can view, share, or embed the dashboard online.

Steps:

- Publish to Tableau Public
- Generate shareable link
- Export charts as images or PDFs (if needed)

Advantages

- **Interactive Dashboards:** Users can filter by house age, renovation, and price for personalized insights.
- **Simplified Analysis:** Visuals make complex housing data easy to understand.
- **Supports Decisions:** Helps real estate teams make pricing and marketing decisions.
- **Web-Friendly:** Dashboards can be published or embedded online.
- **Scalable Design:** Can be extended with new data or integrated with future tools.

Disadvantages

- **Static Data Risk:** Insights depend on updated CSV uploads.
- **Limited Prediction Support:** Tableau is not ideal for advanced ML or forecasting.
- **Initial Learning Curve:** Non-technical users may need guidance.
- **Performance Issues:** Very large datasets may slow performance.

Conclusion

- This project provides a comprehensive view of housing market trends through interactive Tableau dashboards.
- It simplifies raw housing data into insightful visuals for real estate decision-making.
- Stakeholders can explore trends by **sale price**, **renovation year**, **house age**, and more.
- The project enhances **data-driven strategies** for marketing, pricing, and customer targeting.
- With web integration and future expansion possibilities, this solution is both **scalable** and **impactful**.



“Data is a precious thing and will last longer than the systems themselves.”

— Tim Berners-Lee